Jedha

CDSD block 2

Cyril Braguy



Project Steam

Steam's videogames platform

CDSD block 2
Cyril Braguy

Goal of the project

Pitch

- **Steam** is a video game digital distribution service and storefront from Valve with software client launched in 2003 to provide game updates automatically for Valve's games, and expanded to distributing third-party titles in late 2005. Steam offers various features, like digital rights management (DRM), game server matchmaking with Valve Anti-Cheat measures, social networking, and game streaming services. Steam client's functions include game update automation, cloud storage for game progress, and community features such as direct messaging, in-game overlay functions and a virtual collectable marketplace. ## Project 🚧

- You're working for Ubisoft, a French video game publisher. They'd like to release a new revolutionary videogame! They asked you conduct a global analysis of the games available on Steam's marketplace in order to better understand the videogames ecosystem and today's trends.

Goal

- Understand what factors affect the popularity or sales of a video game. And to analyze the video game market globally.

- Adopt different levels of analysis by surveying :

- Analysis at the "macro" level

- Genres analysis

- Platform analysis

Analysis is performed from a dataset available on aS3 bucket by using Databricks and PySpark to conduct this EDA,
Particularly, you'll have to use Databrick's visualisation tool to create the visualizations.

Load dataset

- Extract tags and dicts
- Flatten
- Feature engineerong
- Create special views for analyses

- As the dataset is semi-structured with a nested schema, use getField() and explode()
- There are some text and date fields in this dataset: utilitary functions are used 💡
- Then agregate functions and groupBy are used to conduct segmented analysis.

```
root
 |-- appid: long (nullable = true)
 |-- categories: array (nullable = true)
 |    |-- element: string (containsNull = true)
 |-- ccu: long (nullable = true)
 |-- developer: string (nullable = true)
 |-- discount: string (nullable = true)
 |-- genre: string (nullable = true)
 |-- header_image: string (nullable = true)
 |-- initialprice: string (nullable = true)
 |-- languages: string (nullable = true)
 |-- name: string (nullable = true)
 |-- negative: long (nullable = true)
 |-- owners: string (nullable = true)
 |-- platforms: struct (nullable = true)
 |    |-- linux: boolean (nullable = true)
 |    |-- mac: boolean (nullable = true)
 |    |-- windows: boolean (nullable = true)
 |-- positive: long (nullable = true)
 |-- price: string (nullable = true)
 |-- publisher: string (nullable = true)
```

✓ Sep 21, 2026 (1s) 20

```
df.show(10)
```

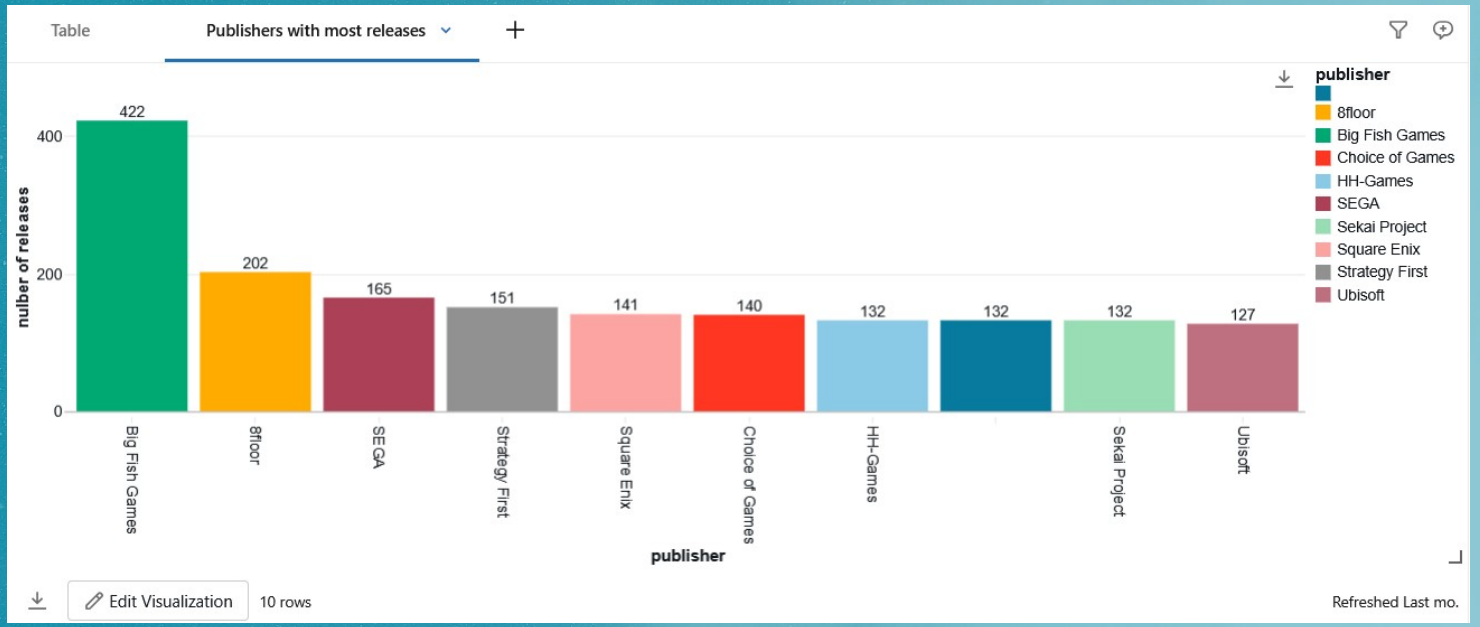
appid	categories	ccu	developer	discount	genre	header_image	initialprice	languages	
10	[Multi-player, Va...	13990	Valve	0	Action	https://cdn.akama...	999	English, French, ...	
1000000	[Single-player, P...	0	IndigoBlue Game S...	0	Action, Adventure...	https://cdn.akama...	999	English, Korean, ...	
1000010	[Single-player, P...	99	NEXT Studios	70	Adventure, Indie,...	https://cdn.akama...	1999	Simplified Chines...	
1000030	[Multi-player, Si...	76	Vertigo Gaming Inc.	0	Action, Indie, Si...	https://cdn.akama...	1999	English	Coo
1000040	[Single-player]	0	DoubleC Games	0	Action, Casual, I...	https://cdn.akama...	199	Simplified Chinese	
1000080	[Multi-player, Si...	3	IndieLeague Studio	60	Action, Adventure...	https://cdn.akama...	1999	Simplified Chines...	
1000100	[Single-player, S...	0	七月九日	0	Adventure, Indie,...	https://cdn.akama...	1299	Japanese, Simplif...	干支セトラ 陽ノ巻 |
1000110	[Multi-player, Si...	0	重庆环游者网络科技	0	Action, Adventure...	https://cdn.akama...	0	English, Simplifi...	Jumping
1000130	[Single-player, S...	0	Simon Codrington	0	Casual, Indie	https://cdn.akama...	299	English	
1000280	[Single-player]	0	Villain Role	0	Indie, RPG	https://cdn.akama...	1399	English, Simplifi...	Tow

only showing top 10 rows

Analysis at the "macro" level

Which publisher has released the most games on Steam?

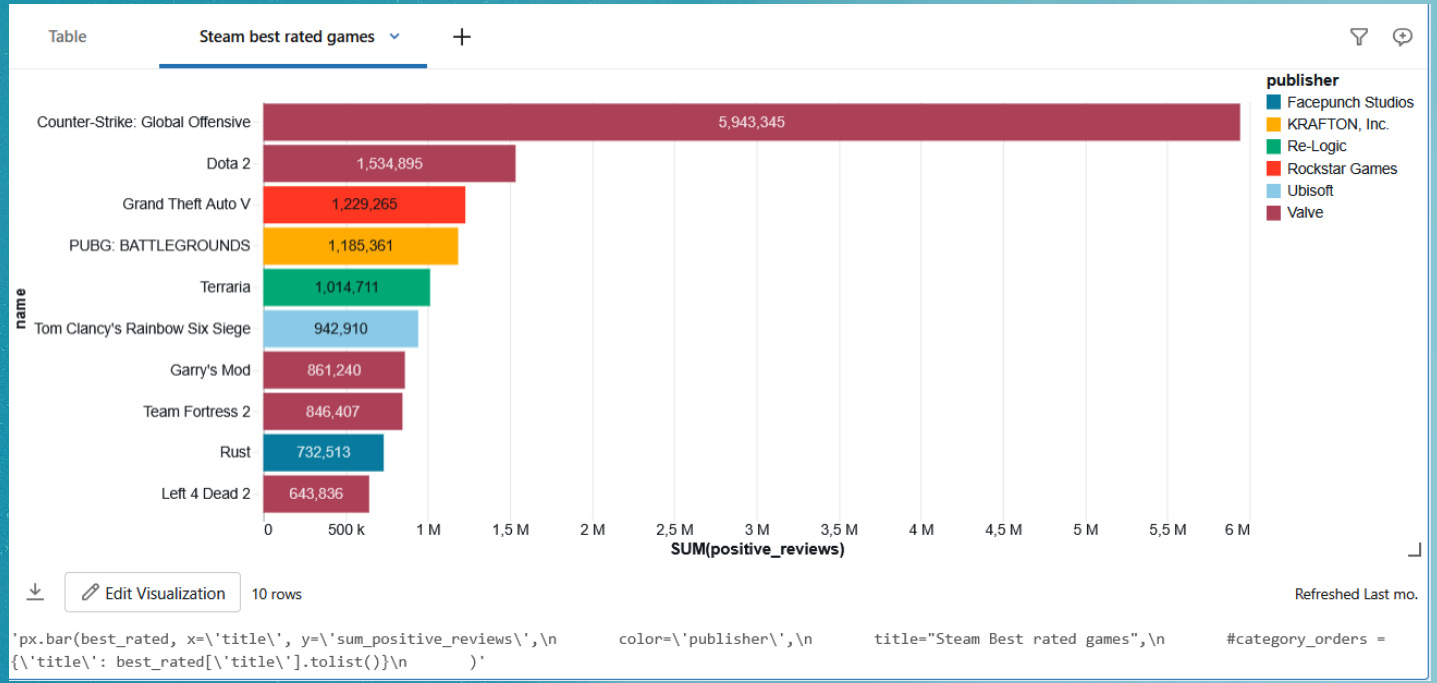
- Big Fish games
- 8floor
- SEGA
- …



Analysis at the "macro" level

What are the best rated games?

- Counter strike global offensive



Analysis at the "macro" level

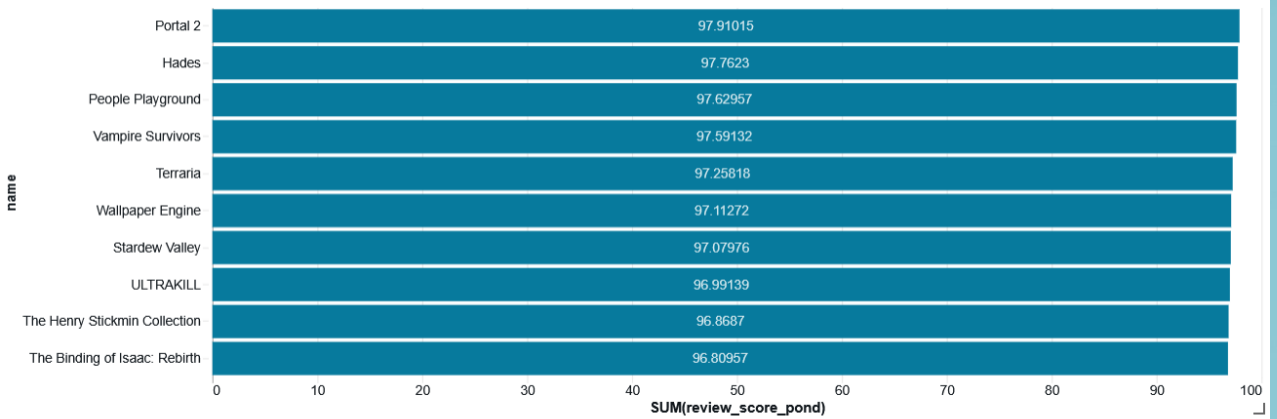
what are the best scored games

```
print(f"Top 5 Best ponderated-scored games")
display(best_rated2.limit(10))
```

best_rated2: pyspark.sql.connect.dataframe.DataFrame = [publisher: string, name: string ... 3 more fields]

Top 5 Best ponderated-scored games

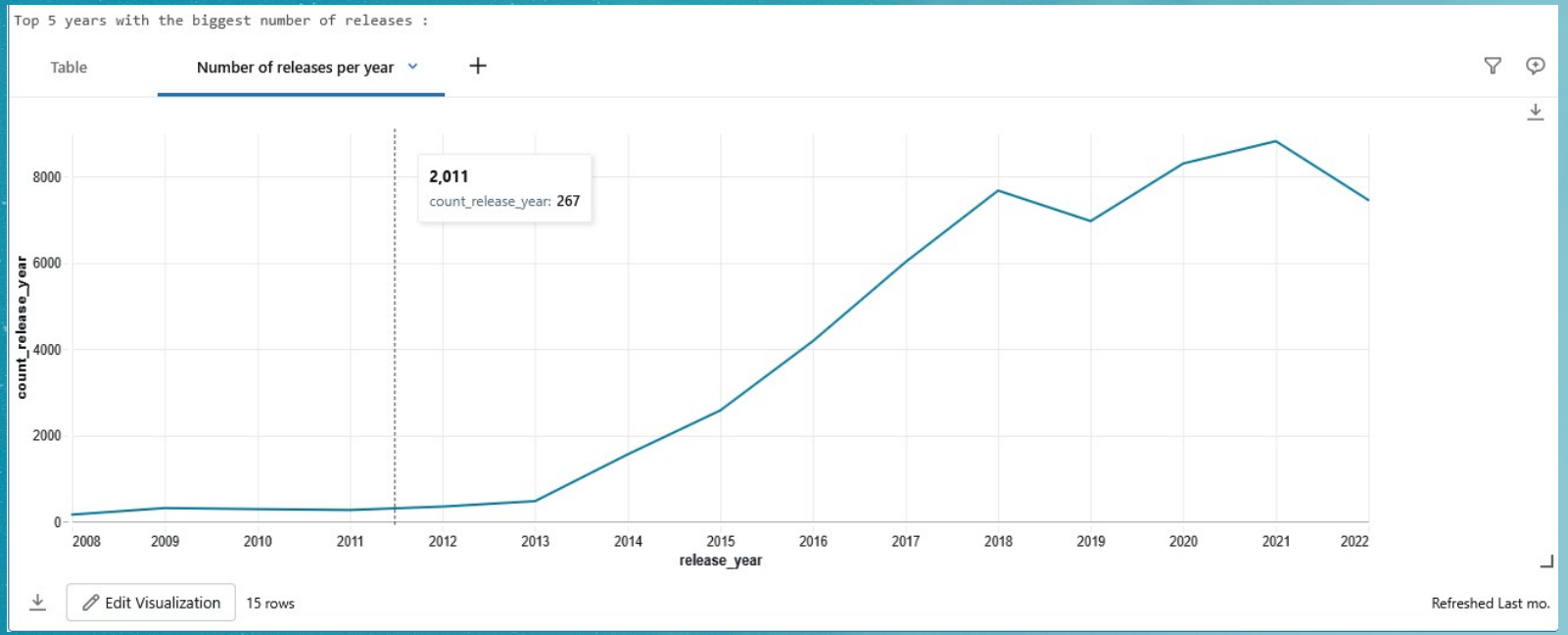
Table | Visualization 1



Edit Visualization | 10 rows | Refreshed Last mo.

Analysis at the "macro" level

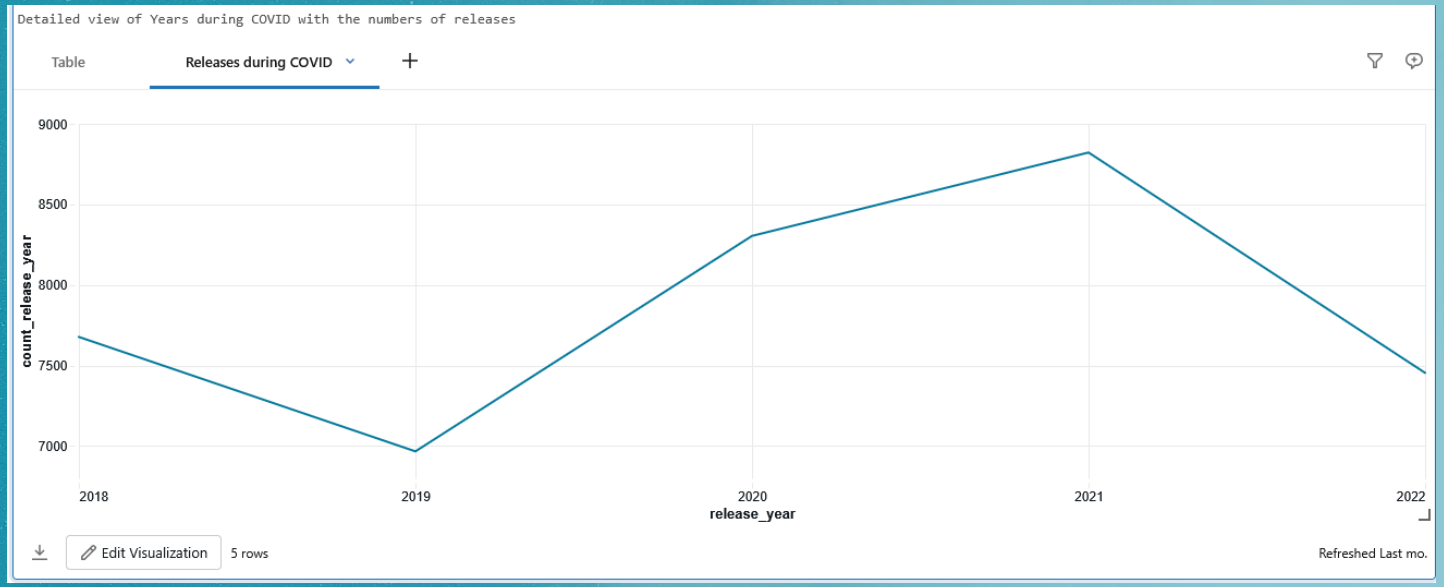
Are there years with more releases?



Analysis at the "macro" level

Were there more or fewer game releases during the Covid, for example? (between 2020 and 2022)

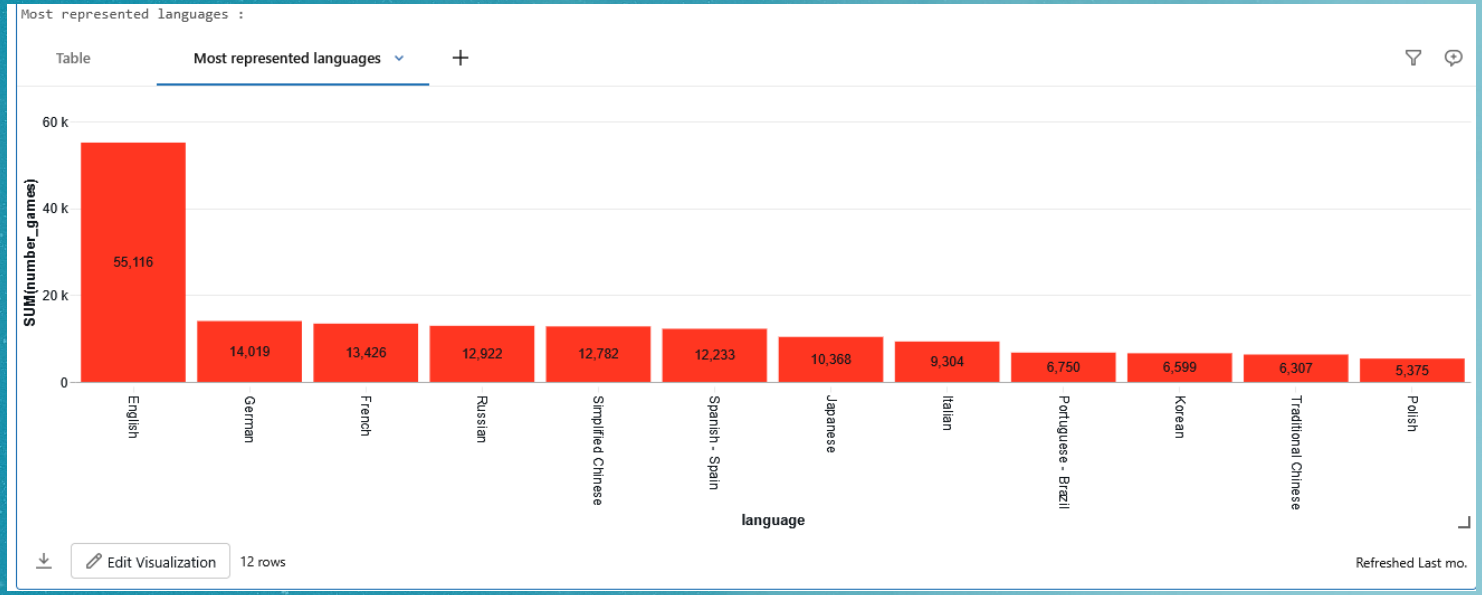
- No reduction of releases during Covid period,
- even an increase is seen



Analysis at the "macro" level

What are the most represented languages?

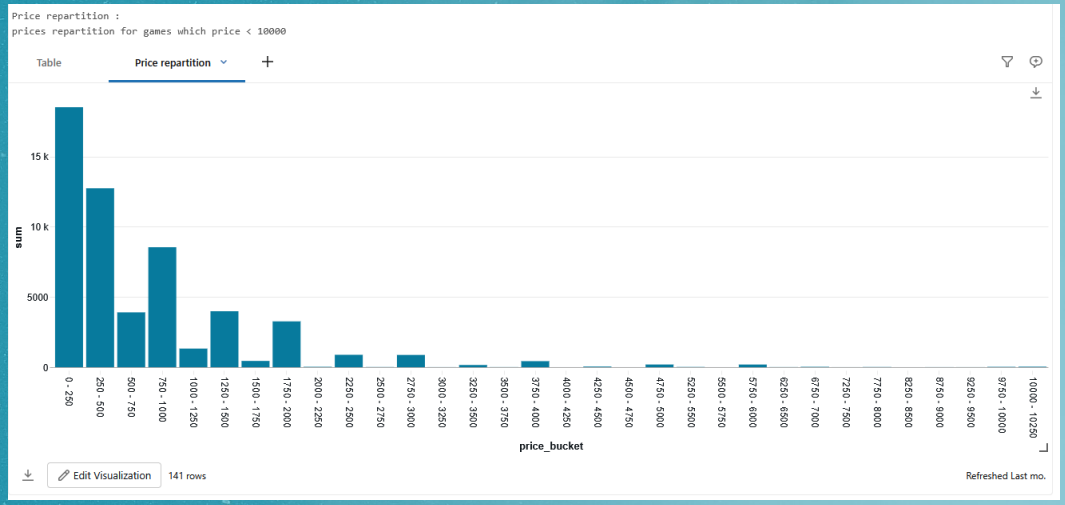
- English at a wide majority
- German
- French
- Russian

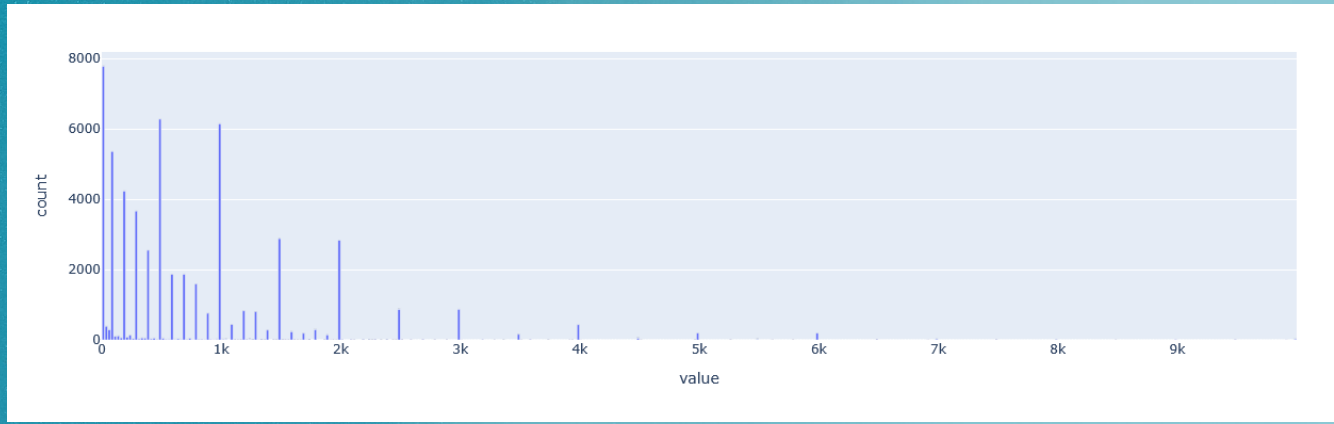


Analysis at the "macro" level

How are the prizes distributed?

- Prices follow a double pattern of 500 and 100
-

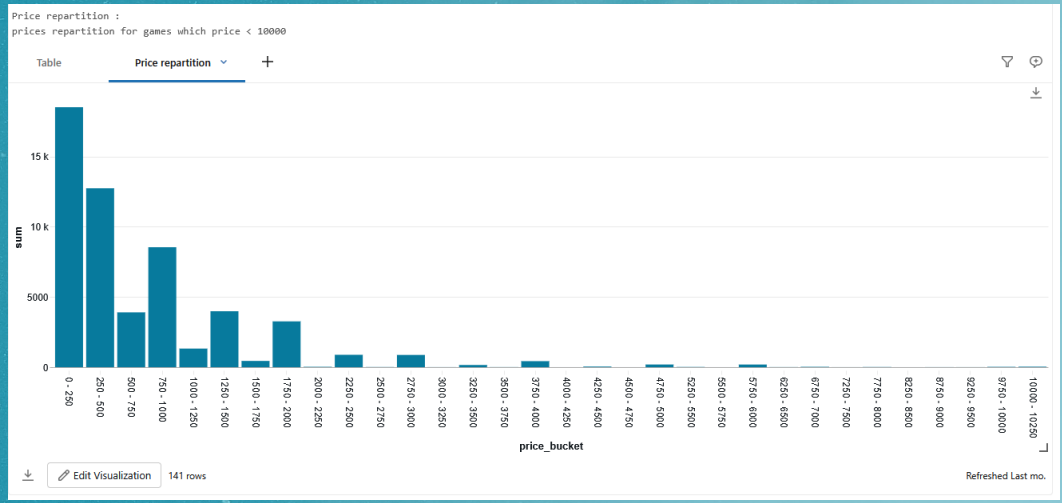


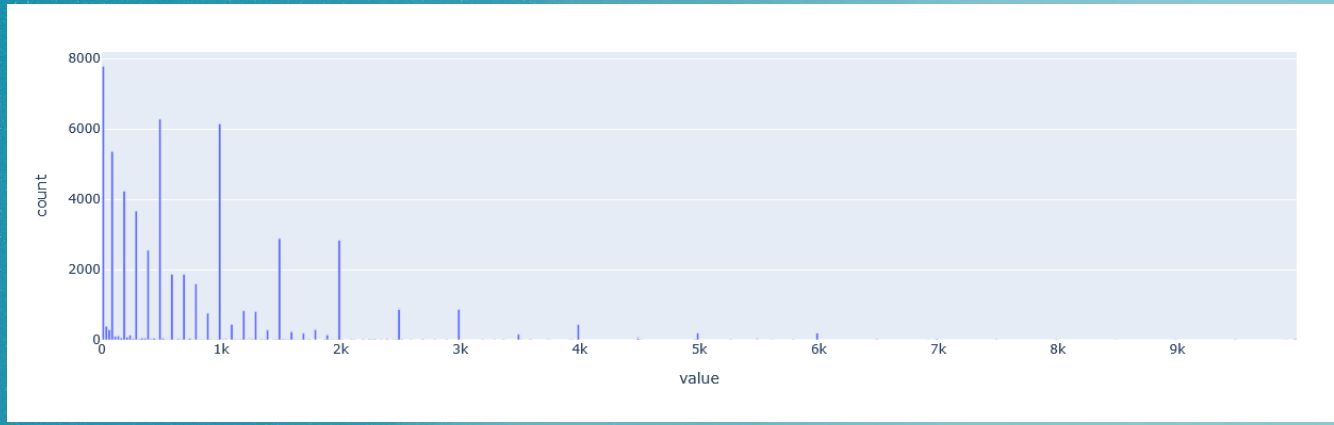


Analysis at the "macro" level

How are the prizes distributed?

- Prices are mainly distributed along 2 families of meshes of 100 and 500
- slowly decreasing
- with a bump around 1000 and 1250

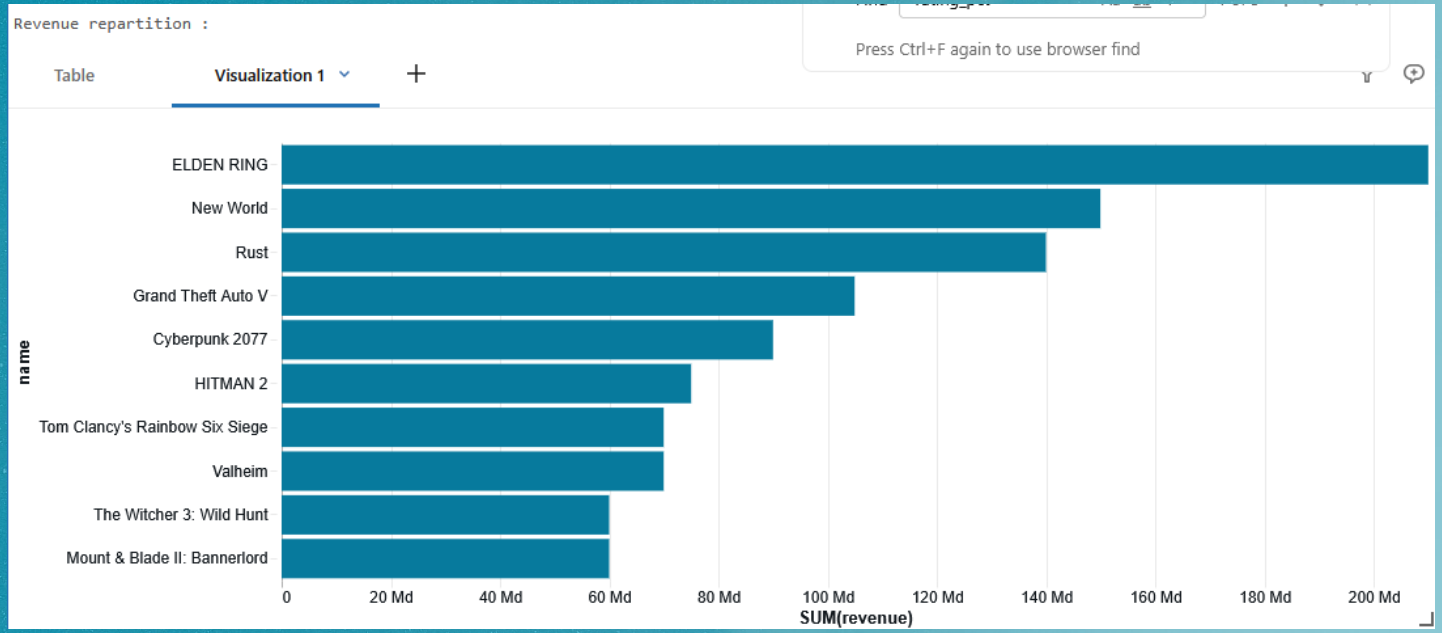




Analysis at the "macro" level

How is revenue distributed

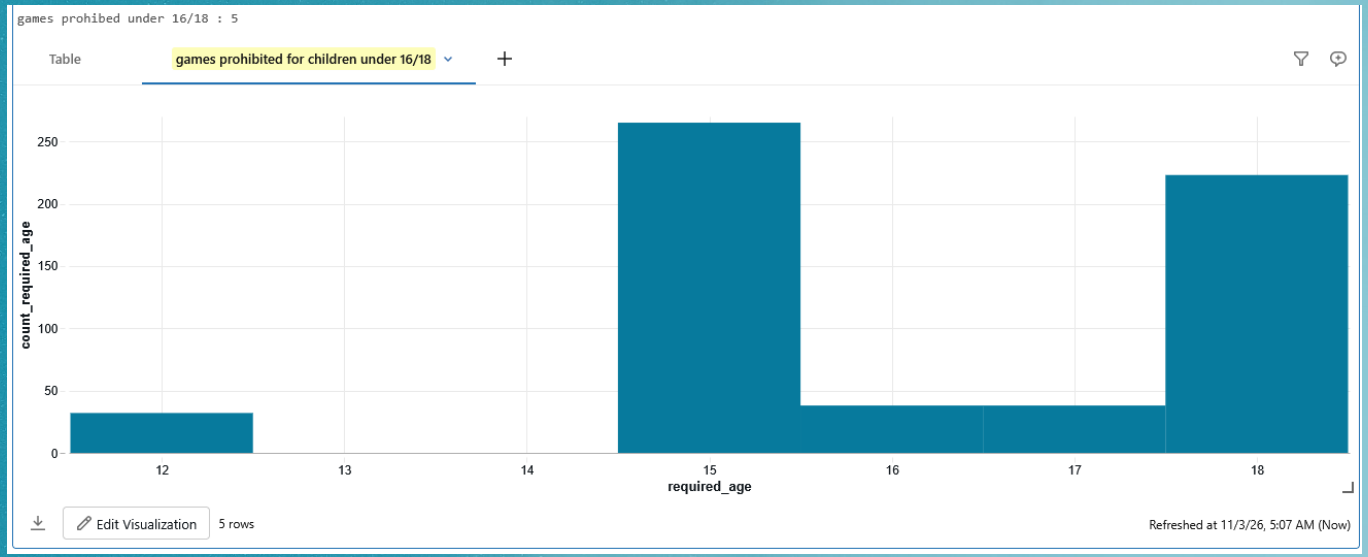
- revenue = price * nb_users_mean
- Eden ring is the winner !



Analysis at the "macro" level

Are there many games prohibited for children under 16/18?

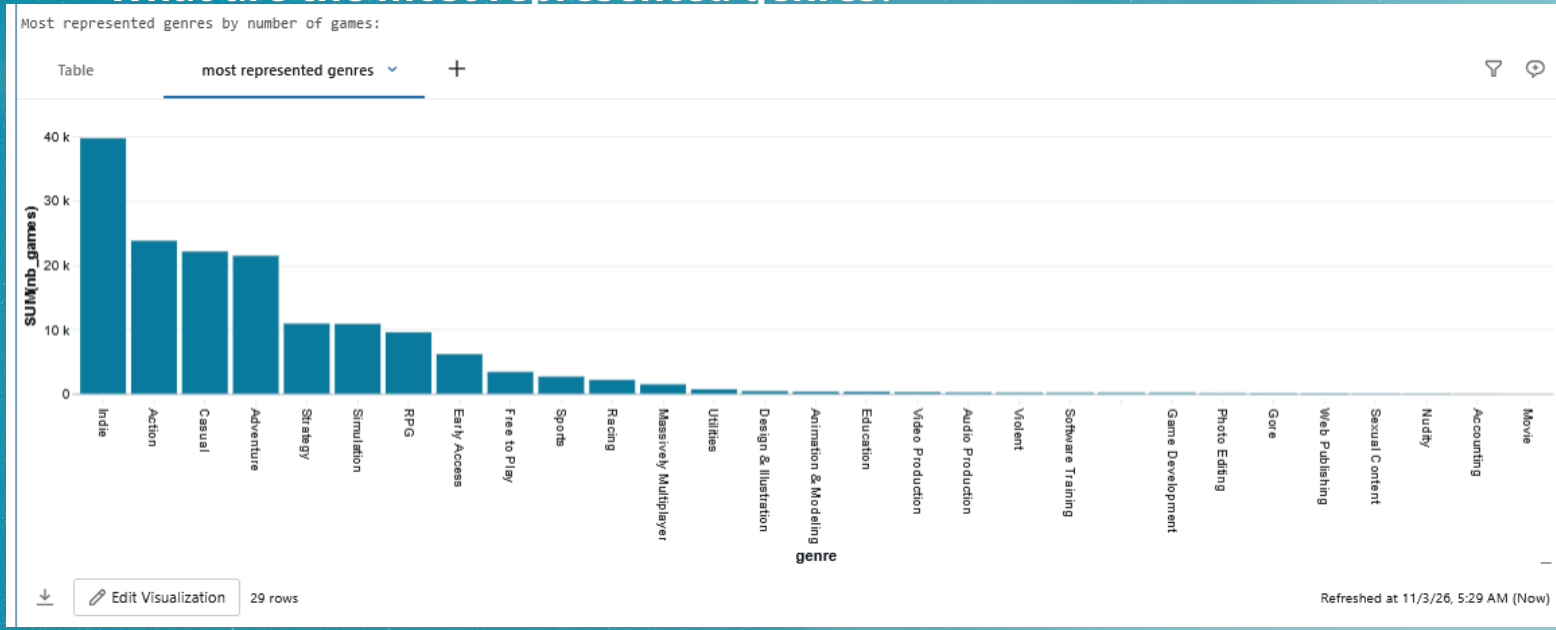
- Main prohibed ages are 16 and 18
- 12 years also are a bit concerned



Genres analysis

What are the most represented genres?

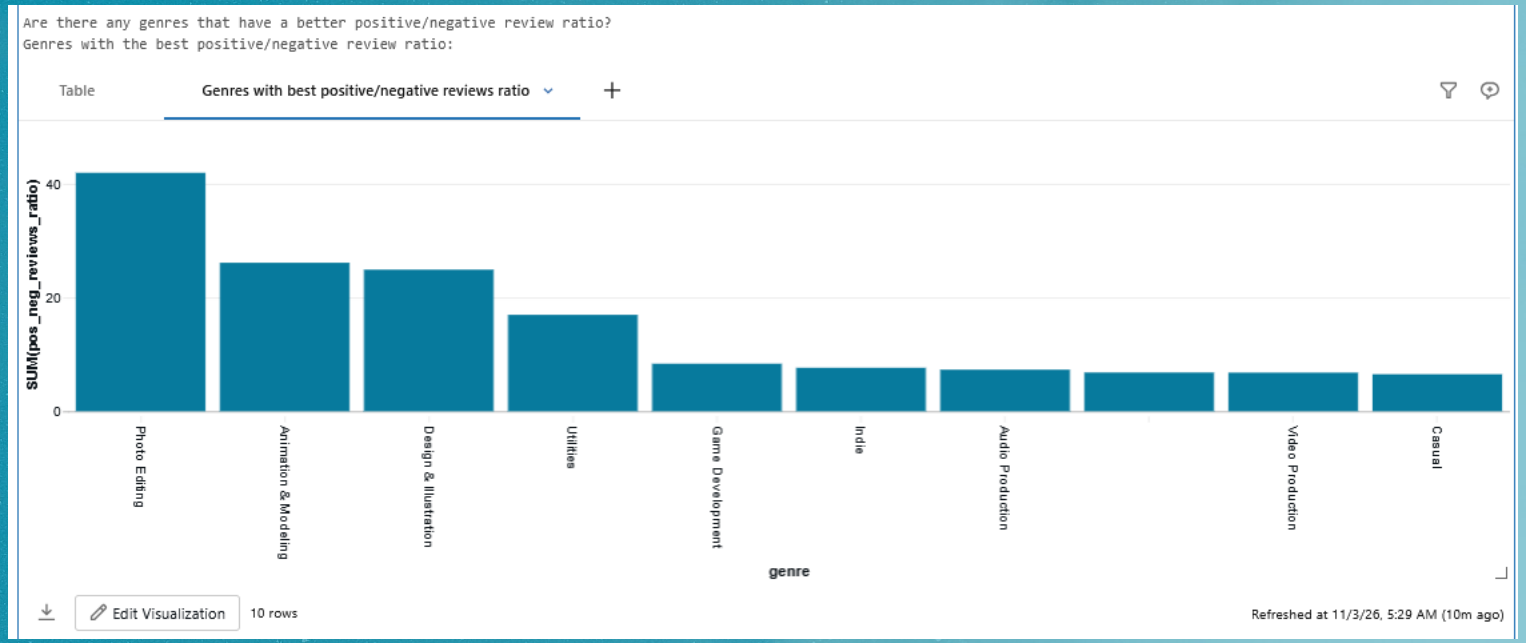
- Most represented genres are Indie,
- then grouped : Action, Casual, Adventure
-



Genres analysis

Are there any genres that have a better positive/negative review ratio?

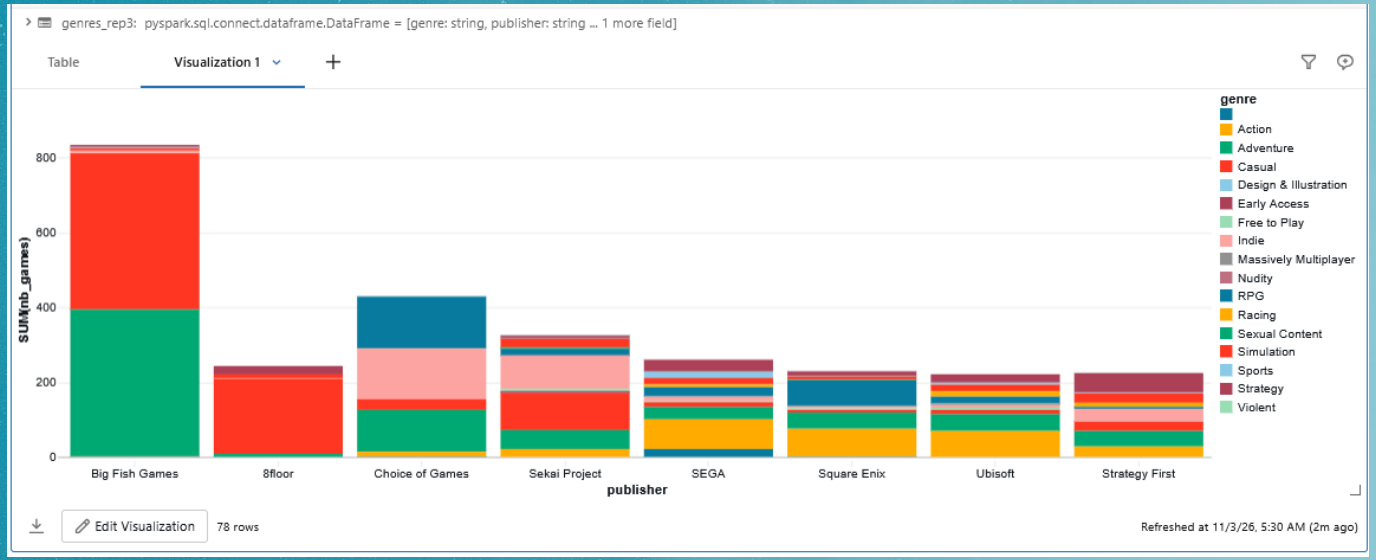
- Best review ratio are for artistic activities, that are not really games :
  - photo editing,
  - animation and modeling,
  - design and illustration



Genres analysis

Do some publishers have favorite genres?

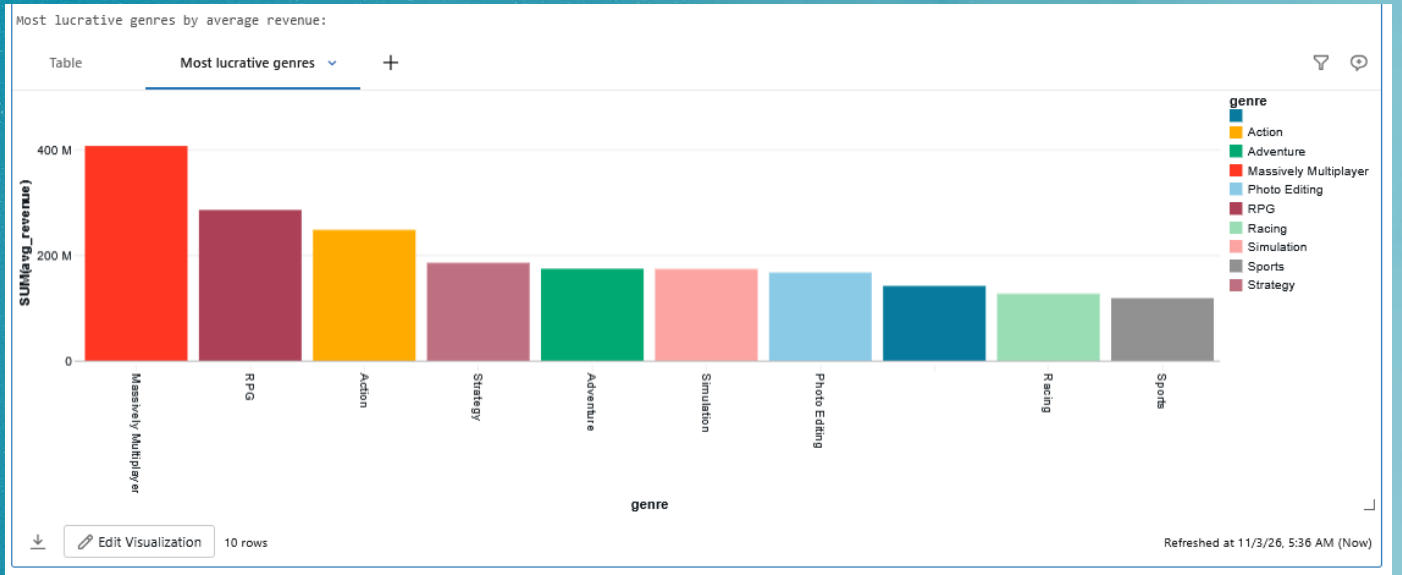
- Definitely we can say yes !
- Big Fish and 8Floor have prefered genres



Genres analysis

What are the most lucrative genres?

- Massive multiplayers
- RPG
- Action
- …
-



3 Platform analysis

Are most games available on Windows/Mac/Linux instead?

- Massive multiplayers
-

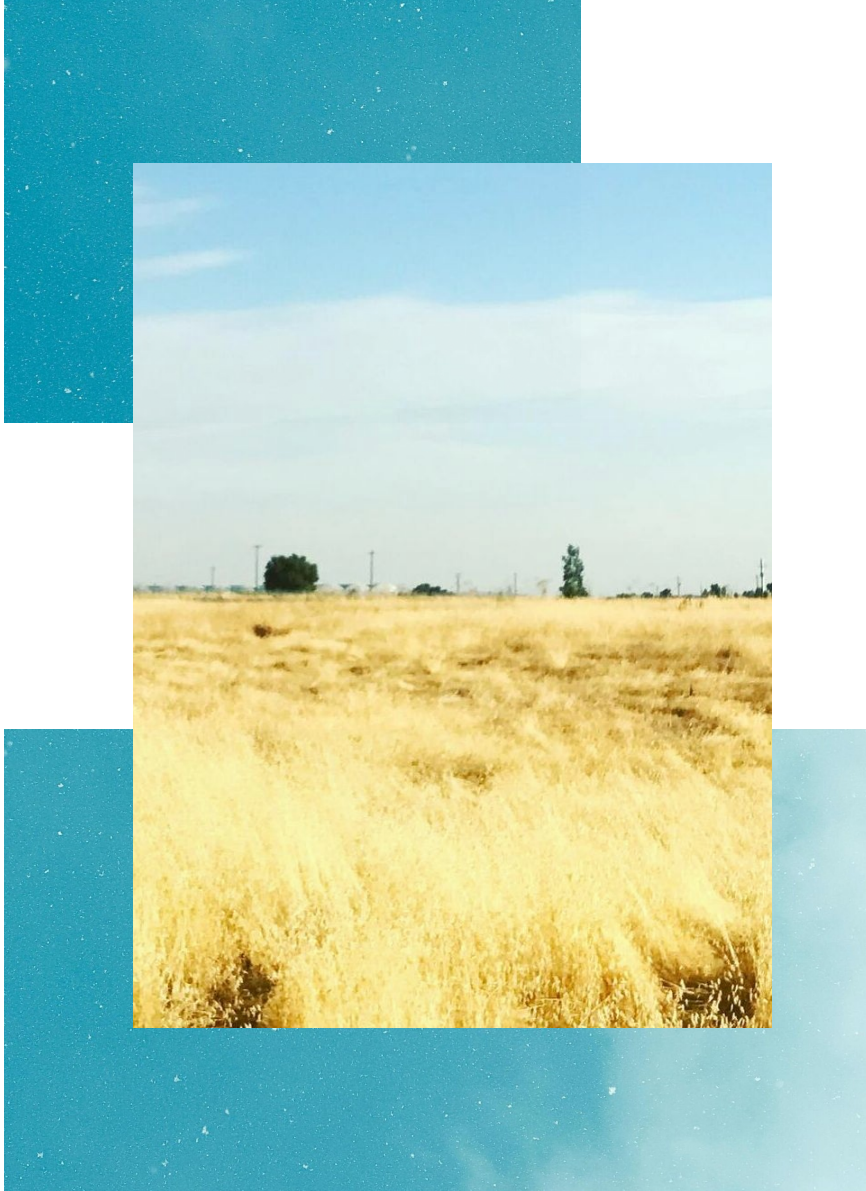
Platform analysis

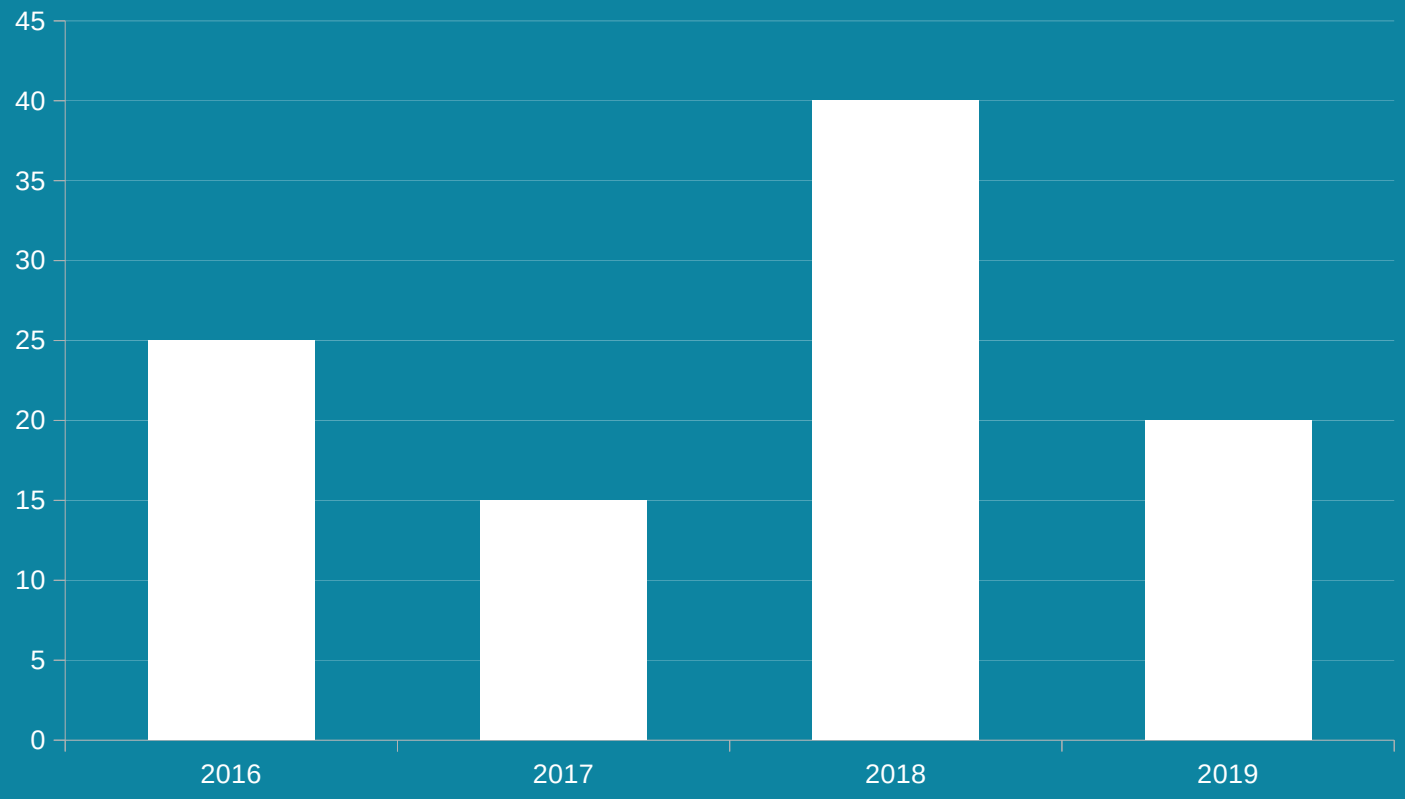
Do certain genres tend to be preferentially available on certain platforms?

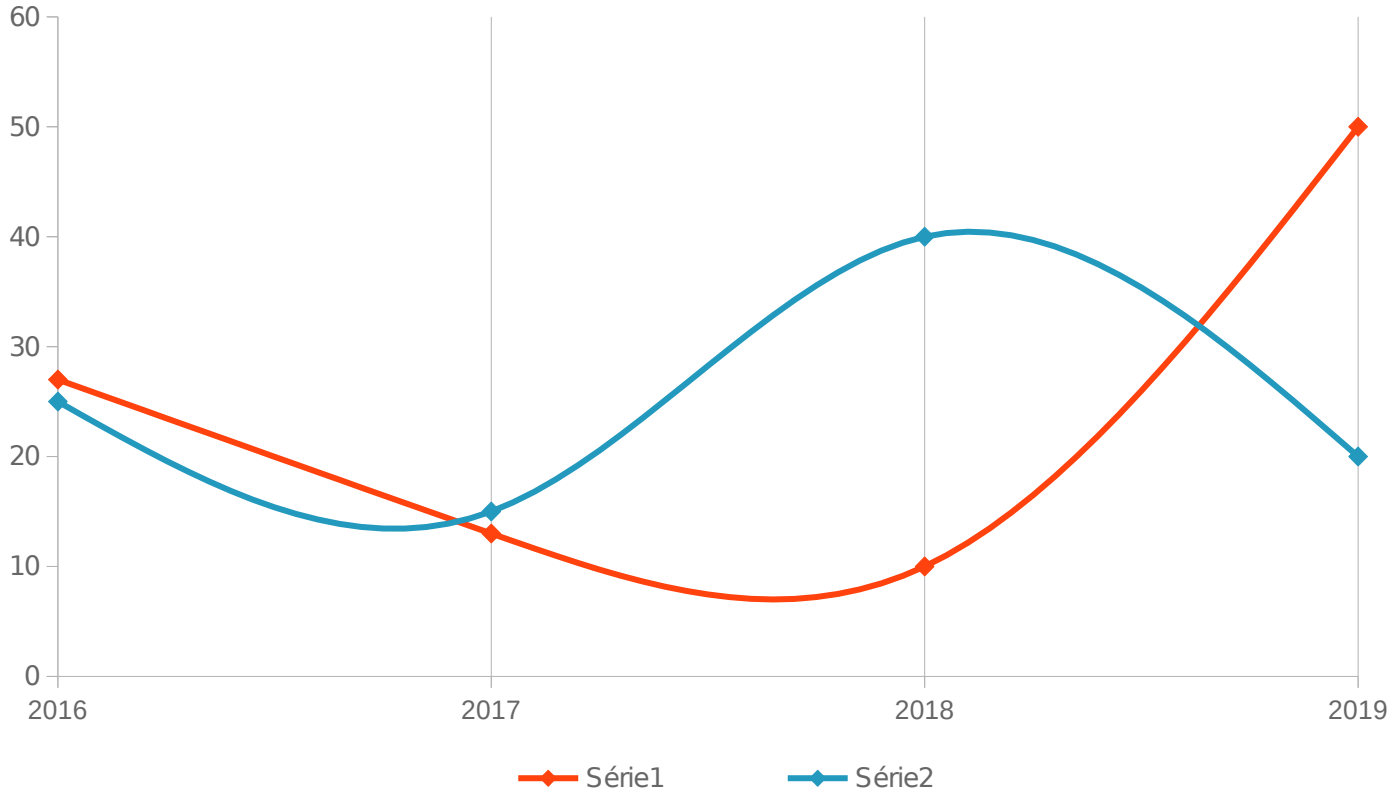
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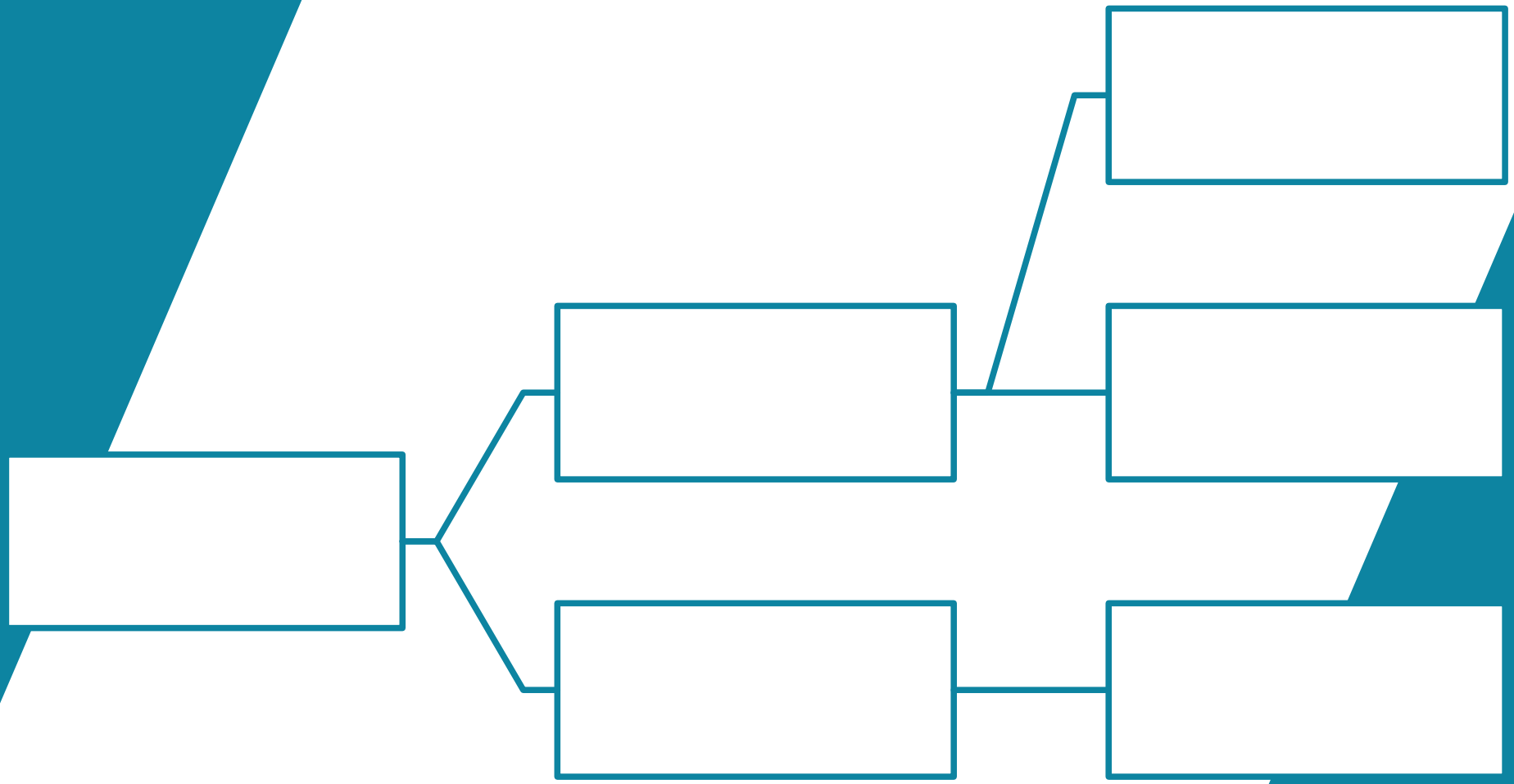







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Lorem Ipsum Dolor 1	M 10	M 18	M 28
Lorem Ipsum Dolor 2	M 23	M 33	M 56
Lorem Ipsum Dolor 3	M 23	M 23	M 46
Lorem Ipsum Dolor 4	M 71	M 17	M 88
Lorem Ipsum Dolor 5	M 34	M 82	M 116
Sit	**M 161**	**M 173**	**M 334**


2016


2020

